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Logical channel management in a wireless communication network

Abstract

A base station may be configured to send a first downlink control information indicating that a wireless transmit/receive unit (WTRU) is to transmit uplink data on a first uplink channel. The base station may be configured to receive a plurality of a same value over a second uplink channel in response to a triggering condition. The second uplink channel may be a control channel. The base station may be configured to sending a second downlink control information in accordance with the plurality of the same value over the second uplink channel. The plurality of the same value may be in response to the first downlink control information providing an insufficient amount to transmit uplink data over the first uplink channel. The base station may be configured to send an indication of logical channels that are subject to triggering of an uplink scheduling request.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 16/852,668 filed Apr. 20, 2020, which is a continuation of U.S. patent application Ser. No. 16/207,795 filed Dec. 3, 2018, which issued as a U.S. Pat. No. 10,630,448 on Apr. 21, 2020, which is a continuation of U.S. patent application Ser. No. 15/581,902 filed Apr. 28, 2017, which issued as U.S. Pat. No. 10,148,405 on Dec. 4, 2018, which is a continuation of U.S. patent application Ser. No. 14/449,819 filed Aug. 1, 2014, which issued as U.S. Pat. No. 9,641,304 on May 2, 2017, which is a continuation of U.S. patent application Ser. No. 12/942,528 filed Nov. 9, 2010, which issued as U.S. Pat. No. 8,797,873 on Aug. 5, 2014, which is a continuation of U.S. patent application Ser. No. 11/840,308 filed Aug. 17, 2007, which issued as U.S. Pat. No. 7,830,835 on Nov. 9, 2010, which claims the benefit of U.S. Provisional Application Ser. No. 60/839,198 filed Aug. 21, 2006, the contents of which are each hereby incorporated by reference herein.

FIELD OF INVENTION

(1) The present invention is related to high speed uplink packet access (HSUPA) wireless communication systems. More particularly, the present invention is related to a method and apparatus for preventing transmission blocking in an HSUPA wireless communication system.

BACKGROUND

(2) The Third Generation Partnership Project (3GPP) Release 6 defines fast control of wireless transmit/receive unit (WTRU) transmissions through Node-B based scheduling in HSUPA. This faster control results in better control of the uplink (UL) noise rise, which allows operation at a higher average UL load without exceeding the threshold, thereby increasing system capacity. In HSUPA, control and feedback occurs through different physical control channels and information elements (IEs).

(3) Node-B commands are conveyed by absolute or relative grant channels, while WTRU feedback is transmitted on an enhanced dedicated physical control channel (E-DPCCH), or “happy bit” within the E-DPCCH, where scheduling information (SI) is appended to the payload. The Node-B commands are expressed by a maximum power ratio over the power of the UL control channel (DPCCH). The happy bit is transmitted within the E-DPCCH along with 2 bits for retransmission sequence number (RSN) and 7 bits for the enhanced transport format combination indication (E-TFCI). All combinations of the 7 E-TFCI bits are defined to mean a specific size of the enhanced transport format combination (E-TFC). The value “0” (7 bits) is defined to mean the transmission of the SI alone. The E-DPCCH is always transmitted along with the enhanced dedicated physical data channel (E-DPDCH) except during compressed mode. Transmission of E-DPCCH alone does not occur.

(4) The WTRU and Node-B are aware how much data can be transmitted for a given power ratio, and this correspondence is controlled by the radio network controller (RNC). Such a scheduled operation is particularly well suited to non-delay-sensitive types of applications, however it may also be used to support more delay-sensitive applications, given the fast resource allocation capabilities.

(5) Under the current standard, data is optionally segmented, and buffered at the radio link control (RLC) layer. The set of possible RLC packet data units (PDU) sizes that are delivered to the medium access control (MAC) layer is configured by radio resource control (RRC) signaling. When segmentation takes place, generally the sizes of the PDUs are configured to be of the order of several hundreds of bits to avoid excessive overhead and obtain good coding performance.

Currently, there is no further segmentation at the MAC layer. Accordingly, when a new transmission takes place, an integer number of PDUs, including zero, must be sent.

(6) Since it is not possible to send out a fraction of an RLC PDU, a certain minimum instantaneous bit rate for the WTRU transmission is imposed. For instance, if the PDU size is 320 bits and the transmission time interval (TTI) is 2 milliseconds (ms), the instantaneous bit rate needs to be at least 160 kilobits per second (kbps), without accounting for MAC overhead. Such an instantaneous bit rate translates into a certain minimum transmission power ratio, under which no RLC PDUs can be sent.

(7) During scheduled operation, WTRU transmissions from a given MAC-d flow can be completely interrupted, or “blocked,” if the granted power ratio falls under the minimum required to transmit the RLC PDU at the head of buffer. Such a situation may occur out of the control of the serving radio link set, (i.e., Node-B) for a number of reasons. For example, the WTRU may have received a non-serving relative grant requesting a decrease of power from another Node-B, the WTRU may have erroneously decoded a relative or absolute grant command from the serving Node-B, or the WTRU may have several different configured RLC PDU sizes on a given MAC-d flow and a bigger than usual RLC PDU size is up for transmission.

(8) When such a situation occurs, the WTRU cannot transmit until the time it is scheduled to transmit an SI. Until then, and unless the previous SI has been transmitted recently enough for the Node-B to infer that the WTRU buffer is not empty based on its subsequent transmissions, the Node-B has no ability to determine whether transmission stopped because the power ratio fell under the minimum, or simply because the WTRU has nothing to transmit. Accordingly, transmission from the WTRU is delayed until the SI can be transmitted.

(9) This issue imposes a configuration of a small periodicity of SI transmission (T_SIG) for delay-sensitive applications, thereby increasing overhead. Furthermore, even if the Node-B was aware that transmission stopped because the power ratio is too low, when multiple RLC PDU sizes are configured, the Node-B does not know what power ratio to apply to correct the situation. Thus, the Node-B has to find out by trial and error what the correct power ratio is. This results in inefficient resource allocation and/or excessive scheduling delays.

(10) In the current state of the art, transmission of scheduling information (SI) is only allowed under certain conditions such as those described in 3GPP TS 25.321, such as if the user has a grant (power ratio) of zero or has all its processes de-activated and has data to transmit, upon a change of E-DCH serving RLS (base station), or periodically, with a configurable period depending on whether the user has a grant or not. Accordingly, a solution to prevent blocking that would be compatible with the mechanisms defined in the current state of the art may include configuring periodic reporting of the SI with a very low period, such that the SI is transmitted along with almost every transmission of new data. However, overhead may be significantly increased since each SI takes up 18 bits. For instance, assuming a MAC service data unit (SDU) size of 280 bits and a MAC-e header size of 18 bits, this would represent an additional overhead of approximately 6%.

(11) It would therefore be beneficial to provide a method and apparatus for transmission blocking in an HSUPA wireless communication system that is not subject to the limitations of the current state of the art.

SUMMARY

(12) The present invention is related to a method and apparatus for preventing transmission blocking. Transmission of scheduling information (SI) when transmission of a medium access control-d (MAC-d) flow is stopped. The SI is transmitted when the triggering condition is met.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) A more detailed understanding of the invention may be had from the following description of a preferred embodiment, given by way of example and to be understood in conjunction with the accompanying drawings wherein:

(2) FIG. 1 is a functional block diagram of a WTRU and a Node-B, configured in accordance with the present invention;

(3) FIG. 2 is a flow diagram of a method for preventing transmission blocking in an HSUPA wireless communication system in accordance with the present invention;

(4) FIG. 3 is a flow diagram of a method for preventing transmission blocking in an HSUPA wireless communication system in accordance with another embodiment of the present invention; and

(5) FIG. 4 is a flow diagram of a method for preventing transmission blocking in an HSUPA wireless communication system in accordance with another embodiment of the present invention.

DETAILED DESCRIPTION

(6) When referred to hereafter, the terminology “wireless transmit/receive unit (WTRU)” includes but is not limited to a user equipment (UE), a mobile station, a fixed or mobile subscriber unit, a pager, a cellular telephone, a personal digital assistant (PDA), a computer, or any other type of user device capable of operating in a wireless environment. When referred to hereafter, the terminology “base station” includes but is not limited to a Node-B, a site controller, an access point (AP), or any other type of interfacing device capable of operating in a wireless environment.

(7) FIG. 1 is a functional block diagram **100** of a WTRU **110** and NB **120** configured in accordance with the present invention. As shown in FIG. 1, the WTRU **110** is in communication with the NB **120** and both are configured to perform a method for preventing transmission blocking in an HSUPA wireless communication system in accordance with the present invention.

(8) In addition to the components that may be found in a typical WTRU, the WTRU **110** includes a processor **115**, a receiver **116**, a transmitter **117**, and an antenna **118**. The processor **115** is configured to perform a method for preventing transmission blocking in an HSUPA wireless communication system in accordance with the present invention. The receiver **116** and the transmitter **117** are in communication with the processor **115**. The antenna **118** is in communication with both the receiver **116** and the transmitter **117** to facilitate the transmission and reception of wireless data.

(9) In addition to the components that may be found in a typical Node-B, the NB **120** includes a processor **125**, a receiver **126**, a transmitter **127**, and an antenna **128**. The processor **115** is configured to perform a method for preventing transmission blocking in an HSUPA wireless communication system in accordance with the present invention. The receiver **126** and the transmitter **127** are in communication with the processor **125**. The antenna **128** is in communication with both the receiver **126** and the transmitter **127** to facilitate the transmission and reception of wireless data.

(10) FIG. 2 is a flow diagram of a method **200** for preventing transmission blocking in an HSUPA wireless communication system in accordance with the present invention. In the present embodiment of the present invention, new conditions for the transmission of the SI are created. In step **210**, a trigger condition for transmitting an SI is detected. For example, the transmission of the SI alone may occur when the transmission of any, or in a specifically defined, MAC-d flow is stopped because the current non-zero grant is smaller than the minimum required to transmit the next MAC SDU, or RLC PDU, of the particular MAC-d flow. The trigger condition, in this case, may occur when it is not possible to transmit a single PDU of a given MAC-d flow. Preferably, a MAC-d flow is a group of logical channels that may be identified, or specified, with an index.

(11) Once the trigger condition is determined, a particular WTRU **110** transmits the SI (step **220**). This transmission may occur once when the triggering condition is met and periodically thereafter,

(e.g. over a configurable period), or the transmission may occur at any time the triggering condition occurs. Additionally, the list of MAC-d flows subject to triggering the transmission of SI due to blocking may be signaled by higher layers, as well as the configured periodicity of transmission once the condition is met.

(12) FIG. 3 is a flow diagram of a method **300** for preventing transmission blocking in an HSUPA wireless communication system in accordance with another embodiment of the present invention. In step **310**, a trigger condition is detected. Preferably, the trigger condition met in step **310** is substantially similar to the trigger conditions described in step **210** of method **200** above. However, unlike the method **200**, when the trigger condition is detected in step **310**, instead of transmitting the SI, nothing is transmitted on the E-DPDCH and all 10 bits of the E-DPCCH are set to a value of zero "0" (step **320**).

(13) In effect, this corresponds to the same setting as for the initial transmission of an SI alone, except that the SI is not actually transmitted. An advantage of this technique is that the required transmission power may be lowered further than if the SI is actually transmitted. However, the E-DPCCH should be transmitted at a value high enough for the network to detect that something has been transmitted on the E-DPCCH. Additionally, less information may be available to the network about the status of the buffer in the WTRU **110**.

(14) FIG. 4 is a flow diagram of a method **400** for preventing transmission blocking in an HSUPA wireless communication system in accordance with another embodiment of the present invention. In the present embodiment of the invention, improved feedback indicating minimum power ratio or MAC SDU size is utilized.

(15) In the current state of the art, the possible MAC SDU sizes, or equivalently RLC PDU sizes, are configured upon radio bearer setup or reconfiguration through RRC signaling. The NB **120** is also aware of the PDU sizes through NB application part (NBAP) signaling. The power ratio grant required to transmit an E-TFC (MAC-e PDU) of a certain size is known by the WTRU **110**, NB **120** and RNC, and any modification is signaled through RRC/NBAP signaling. Thus, using information available with the current standard, the NB **120** could determine what power ratio is required to transmit an E-TFC containing a single RLC PDU for each configured RLC PDU size.

(16) By utilizing the signaling defined in the current standard, the NB **120** may reduce the frequency of occurrence of the issue by never signaling a power ratio to the WTRU **110** that is lower than what is required to transmit the largest RLC PDU size among the RLC PDU sizes configured. It may, however, still be possible that the WTRU **110** blocks transmission because it received a "down" non-serving relative grant or because it misinterpreted a serving grant. The NB **120** should assume the largest RLC PDU because it is not aware of the size of the next RLC PDU in line for transmission at the WTRU's side. As soon as there is more than one RLC PDU size configured, the NB **120** over-allocates resources for the WTRU **110** whenever it is using one of the smaller RLC PDU sizes.

(17) Accordingly, a new type of control information may be signaled by the WTRU **110** to the NB **120**, so that the NB **120** may be aware of the minimum power ratio that should be granted to the WTRU **110** with respect to the size of an upcoming RLC PDU buffered for transmission. This information, preferably, may be referred to as the minimum grant information (MGI).

(18) In step **410** of the method **400**, the MGI is set. The setting of the MGI may be accomplished in a number of ways. For example, the MGI may be set to the size of the next RLC PDU in line for transmission, (i.e., after the current E-TFC is transmitted), on one of the highest priority MAC-d flows having data in its buffer, or on specific MAC-d flows that may be configured by RRC signaling. Additionally, the MGI may be set according to the size of the largest buffered RLC PDU of the highest priority MAC-d flow. Also, the MGI may be set according to the size of the largest buffered RLC PDU of the highest priority MAC-d flow, or on specific MAC-d flows, expected to be transmitted by a certain delay with the current grant and number of active processes. The delay may be also be configured by RRC signaling.

(19) After determining that the MGI should be transmitted, the MGI is then encoded (step **420**). An “upcoming RLC PDU” may be used to describe an RLC PDU that has its size used for setting the value of fields of the MGI. The MGI may then be encoded according to a variety of methods. For example, the MGI may be encoded to consist of 5 bits and represent a power ratio with a mapping, such as bit mapping, similar to that found in the 3GPP TS 25.212 specification. In this case, the signaled power ratio will be the smallest value that allows transmission of the upcoming RLC PDU.

(20) Alternatively, the MGI may be encoded by a smaller number of bits and represent a power ratio. However, in this case, the mapping may be different and have a lower granularity than the mapping found in the 3GPP TS 25.212 specification. For example, the MGI may be encoded with less than 5 bits as described above. Additionally, the mapping might be pre-established.

(21) In another alternative, the MGI may consist of a variable number of bits depending on how many potential RLC PDU sizes should be represented. For instance, in case there are 4 configured RLC PDU sizes, 2 MGI bits would be required, and each combination would represent a specific RLC PDU size. It should be noted that not all configured RLC PDU sizes necessarily need to be mapped. Accordingly, in case only a subset of RLC PDU sizes is mapped, the WTRU **110** sets the MGI according to the smallest RLC PDU size larger than the upcoming RLC PDU.

(22) The MGI is then transmitted by the WTRU **110** (step **430**). The triggering of the MGI may occur in one of several ways. For example, the MGI may be transmitted once whenever its value would change in accordance with the MGI settings. Also, the MGI may be transmitted at each of a particular number (N) of new MAC-e transmissions, where N is configurable by the radio resource controller (RRC). Additionally, it may be required that two consecutive transmissions of the MGI are separated by a delay of at least a particular number (M) transmission time intervals (TTI), where M is also configurable by the RRC.

(23) Once transmitted, the MGI is received and decoded by the NB **120** (step **440**) preferably at the same time as a MAC-e PDU, and the NB **120** makes adjustments based upon the MGI (step **450**). Preferably, the NB **120** adjusts the power ratio to enable transmission of the upcoming RLC PDU buffered for transmission.

(24) In another embodiment of the present invention, the data rate is managed through the use of a scheduling grant. In this embodiment, the transmission is allowed for a minimum number of PDUs (N.sub.min) of the MAC-d flow for every new MAC-e transmission, without regard to the data rate imposed by the power ratio and without regard to the size of the PDU or PDUs.

(25) Under the current 3GPP standard, (e.g., TS 25.309 Release 6), a MAC-d flow is managed through either non-scheduled transmissions or scheduled grants, but not both. Utilizing non-scheduled transmissions for a given MAC-d flow would overcome problems in the current art for this MAC-d flow, at the expense of a loss of control over the amount of interference generated by this flow.

(26) However, in the present hybrid scheduled/non-scheduled embodiment of the invention, the benefit of scheduling grants in terms of noise rise stability is maintained while ensuring that transmission is never completely blocked due to the granted power ratio falling under the threshold for single PDU transmission. The N.sub.min that is allowed for a new transmission may be set through RRC signaling.

(27) If the required power ratio to transmit the N.sub.min PDUs is higher than the current grant, several options may be employed. Preferably, the power ratio is allowed to increase above the current grant to support the transmission of the PDUs. However, the power ratio may also stay at the current grant, with the WTRU **110** selecting the minimum E-TFC that can support the N.sub.min PDUs. Since more data is transmitted in this scenario for the same power, more hybrid automatic repeat request (HARQ) retransmissions for this MAC-e PDU will be required.

(28) For example, assuming that a MAC-d flow has two configured RLC PDU sizes, 300 and 600 bits, a minimum power ratio required to transmit a MAC-e PDU may be assumed to be

(47/15).sup.2 if it contains 2 RLC PDUs of 300 bits and (53/15).sup.2 if it contains a single RLC PDU of 600 bits. In a scenario where the PDU size of 300 bits is transmitted most of the time and the 600 bits is encountered infrequently, the power ratio granted to the WTRU **110** might be maintained at (53/15).sup.2 on the activated HARQ processes for WTRU **110**. When an RLC PDU of 600 bits shows up at the head of the buffer, under the current standard with the MAC-d flow managed by scheduling grant, the transmission would block. With the hybrid non-scheduled/scheduled solution of the present embodiment of the invention, the WTRU **110** would be allowed to transmit its MAC-e PDU containing the RLC PDU of 600 bits and transmission would not be interrupted. For this MAC-e transmission, either the interference might be slightly higher than planned or there might be a higher probability of more HARQ retransmissions, depending on whether the power ratio is allowed to increase above the current grant or not.

(29) Although the features and elements of the present invention are described in the preferred embodiments in particular combinations, each feature or element can be used alone without the other features and elements of the preferred embodiments or in various combinations with or without other features and elements of the present invention. The methods or flow charts provided in the present invention may be implemented in a computer program, software, or firmware tangibly embodied in a computer-readable storage medium for execution by a general purpose computer or a processor. Examples of computer-readable storage mediums include a read only memory (ROM), a random access memory (RAM), a register, cache memory, semiconductor memory devices, magnetic media such as internal hard disks and removable disks, magneto-optical media, and optical media such as CD-ROM disks, and digital versatile disks (DVDs).

(30) Suitable processors include, by way of example, a general purpose processor, a special purpose processor, a conventional processor, a digital signal processor (DSP), a plurality of microprocessors, one or more microprocessors in association with a DSP core, a controller, a microcontroller, Application Specific Integrated Circuits (ASICs), Field Programmable Gate Arrays (FPGAs) circuits, any other type of integrated circuit (IC), and/or a state machine.

(31) A processor in association with software may be used to implement a radio frequency transceiver for use in a wireless transmit receive unit (WTRU), user equipment (UE), terminal, base station, radio network controller (RNC), or any host computer. The WTRU may be used in conjunction with modules, implemented in hardware and/or software, such as a camera, a video camera module, a videophone, a speakerphone, a vibration device, a speaker, a microphone, a television transceiver, a hands free headset, a keyboard, a Bluetooth® module, a frequency modulated (FM) radio unit, a liquid crystal display (LCD) display unit, an organic light-emitting diode (OLED) display unit, a digital music player, a media player, a video game player module, an Internet browser, and/or any wireless local area network (WLAN) module.

Claims

1. A method, implemented by a base station, the method comprising: sending a first downlink control information indicating that a wireless transmit/receive unit (WTRU) is to transmit uplink data on a first uplink channel; sending an indication of logical channels that are subject to triggering of an uplink scheduling request; receiving a plurality of a same value over a second uplink channel in response to a triggering condition, wherein the triggering condition is based on the indication of logical channels, wherein the second uplink channel is a control channel; and sending a second downlink control information in accordance with the plurality of the same value over the second uplink channel.
2. The method of claim 1, wherein the receiving the plurality of the same value is in response to the first downlink control information providing an insufficient amount to transmit uplink data over the first uplink channel.
3. The method of claim 1, further comprising receiving, for a logical channel, an indication of a

recommended uplink resource.

4. The method of claim 3, wherein the indication of the recommended uplink resource is associated with a data rate.
 5. The method of claim 3, further comprising transmitting an uplink scheduling resource for the first uplink channel based on the received indication.
 6. The method of claim 1, wherein the receiving of the plurality of the same value is a request for an uplink scheduling resource.
 7. A base station comprising: a processor; a receiver; and a transmitter; wherein: the processor and the transmitter are configured to transmit a first downlink control information indicating that a wireless transmit/receive unit (WTRU) is to transmit uplink data on a first uplink channel; the processor and the transmitter are further configured to send an indication of logical channels that are subject to triggering of an uplink scheduling request; the processor and the receiver are configured to receive, in response to a triggering condition, a plurality of a same value over a second uplink channel, wherein the triggering condition is based on the indication of logical channels, wherein the second uplink channel is a control channel; and the processor and the transmitter are further configured to transmit a second downlink control information in accordance with the plurality of the same value over the second uplink channel.
 8. The base station of claim 7, wherein the reception of the plurality of the same value is in response to the first downlink control information providing an insufficient amount to transmit uplink data over the first uplink channel.
 9. The base station of claim 7, wherein the processor and the receiver are further configured to receive, for a logical channel, an indication of a recommended uplink resource.
 10. The base station of claim 9, wherein the indication of the recommended uplink resource is associated with a data rate.
 11. The base station of claim 9, wherein the processor and the transmitter are further configured to transmit an uplink scheduling resource for the first uplink channel based on the received indication.
 12. The base station of claim 7, wherein the reception of the plurality of the same value is a request for an uplink scheduling resource.
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